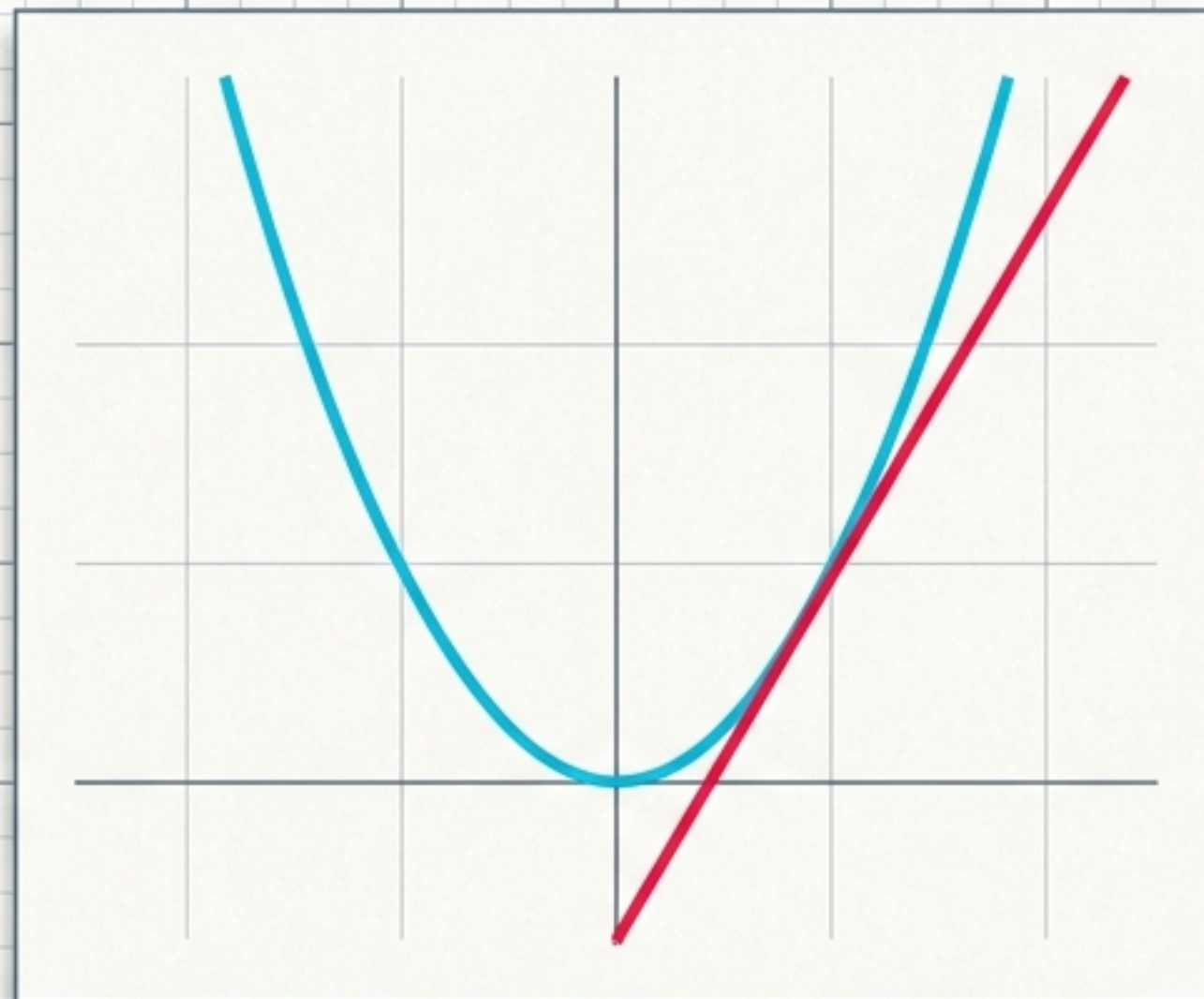
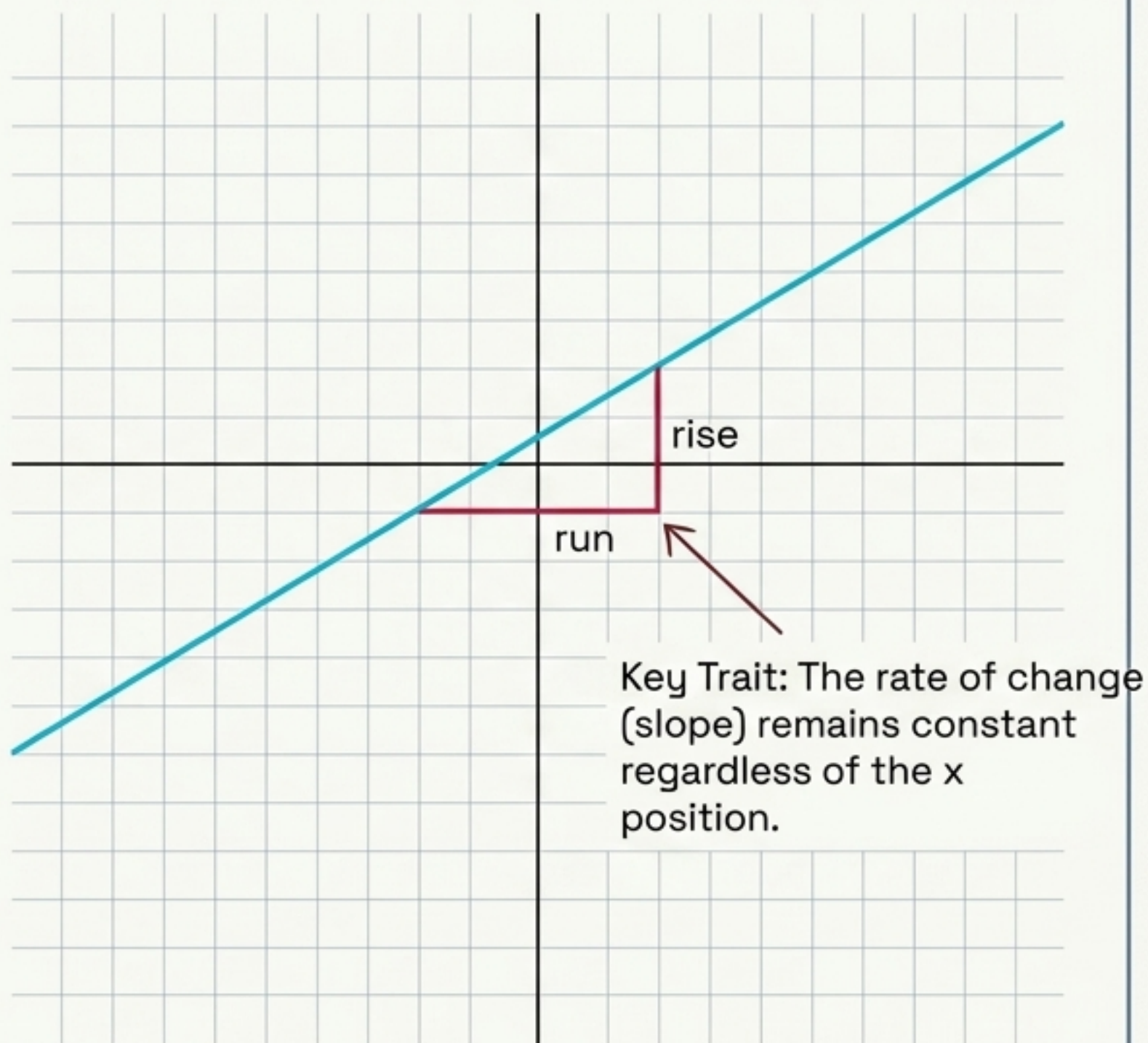




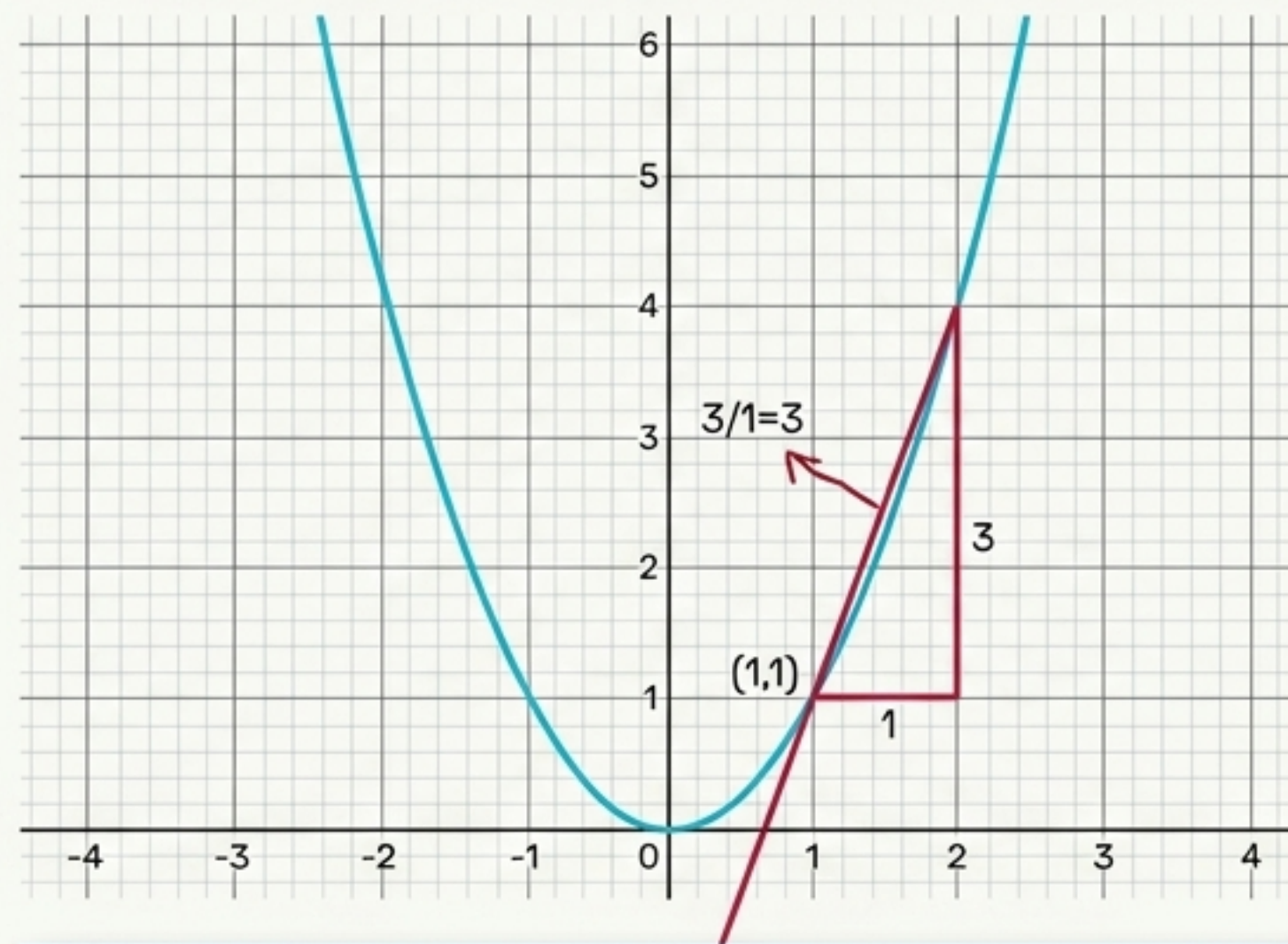
Advanced Derivatives for Machine Learning

A visual reference guide to non-linear rates of change, logarithms, and the Chain Rule.

Linear Equations (Straight Lines)



Non-Linear Equations (Curves)



Key Trait: The rate of change is dynamic. Every single point has a different slope.

- For $x < 0$: Rate of change is decreasing.
- For $x > 0$: Rate of change is increasing.

The Numerical View: Shrinking the Interval

Tracking interval at $x = 1$

$$\Delta x = 1 \rightarrow \text{Slope} = 3$$



$$\Delta x = 0.5 \rightarrow \text{Slope} = 2.5$$



$$\Delta x = 0.25 \rightarrow \text{Slope} = 2.25$$



$$\Delta x = 0.125 \rightarrow \text{Slope} = 2.125$$



$$\Delta x = 0.0001 \rightarrow \text{Slope} = 2.0001$$

As the interval (Δx) gets smaller, the curve mimics a straight line.

Error decreases, and the slope tends exactly to 2.

First Principles: Proving the Quadratic Derivative

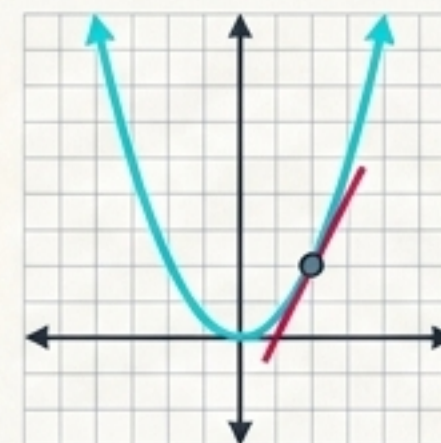
Start: Slope = $\Delta y / \Delta x$ (where $\Delta x \rightarrow 0$)

Expand: $((x + \Delta x)^2 - x^2) / \Delta x$

Unpack: $(x^2 + 2x(\Delta x) + (\Delta x)^2 - x^2) / \Delta x$

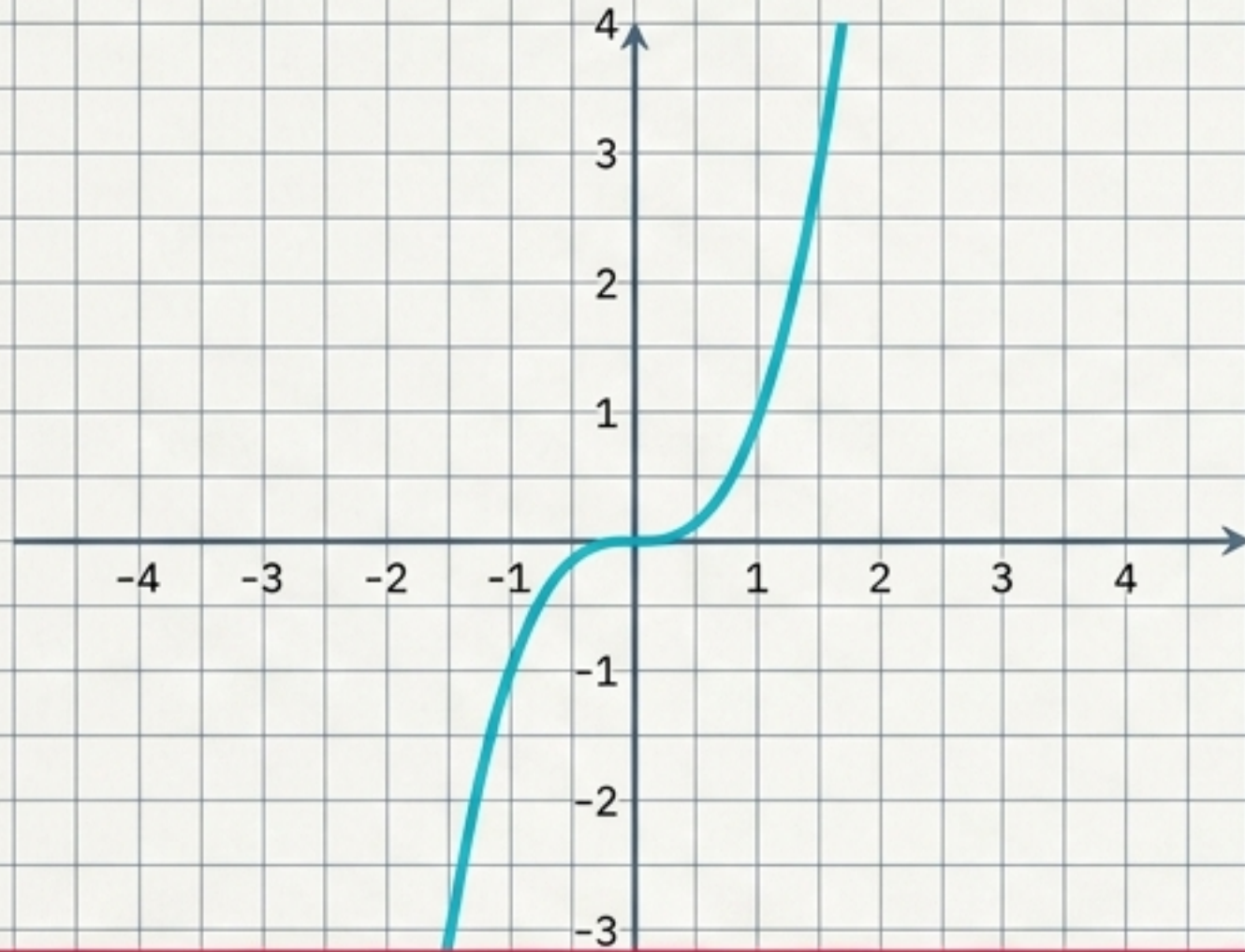
Simplify: $(2x(\Delta x) + (\Delta x)^2) / \Delta x = 2x + \Delta x$

Limit: As $\Delta x \rightarrow 0$, the exact rate of change is **2x**.



For $y = x^2$,
the slope at
any point is
simply **2x**.

The Cubic Function: A Constantly Positive Slope



Observation: As x increases, y strictly increases. The curve never dips.
Conclusion: The derivative must always be positive (or 0).

Function: $y = x^3$

Using First Principles:

$$\frac{(x + \Delta x)^3 - x^3}{\Delta x}$$

Result: $\frac{dy}{dx} = 3x^2$

Insight: Since x is squared in the derivative, the slope calculation will always output a positive number, validating our graphical observation.

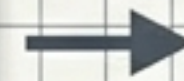
Pattern Recognition: The Universal Power Rule

$$\frac{d}{dx}(x^n) = n * x^{(n-1)}$$

$$y = x^2 \rightarrow \frac{dy}{dx} = 2x$$



$$y = x^3 \rightarrow \frac{dy}{dx} = 3x^2$$

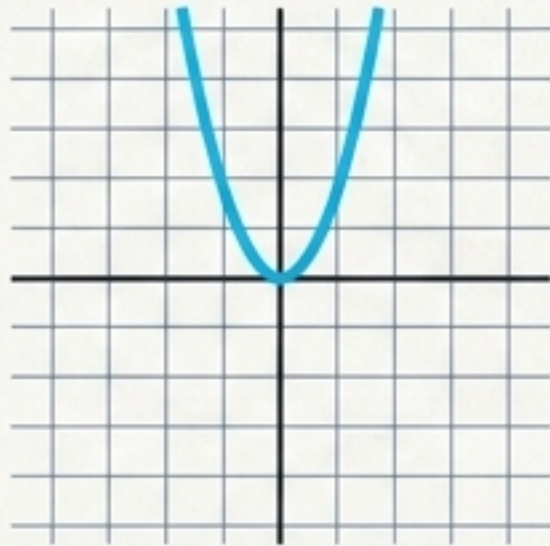


$$y = x^4 \rightarrow \frac{dy}{dx} = 4x^3$$

The exponent drops down to multiply, and the remaining power decreases by one.

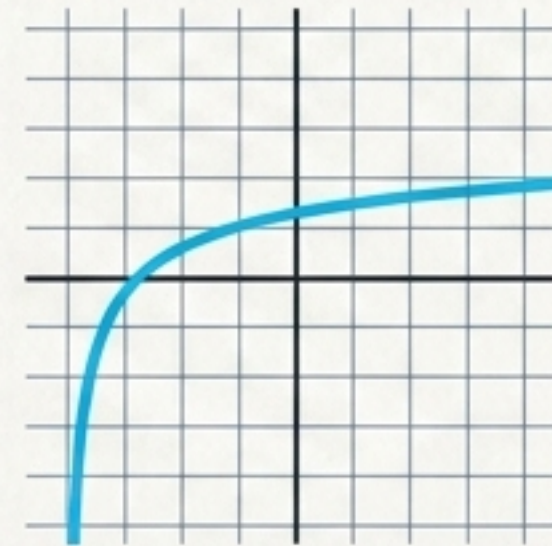
Structural Contrast: Polynomials vs. Logarithms

Polynomials ($y = x^2, x^3$)



- Behavior: Rate of change can rapidly increase or decrease.
- Pace: Changes are steep and aggressive.

Logarithms ($y = \ln(x)$)



- Behavior: Rate of change strictly increases, but does so incredibly slowly.
- Intuition: Logarithms answer 'what power gives this output?'. It requires massive changes in x to yield tiny changes in y .

The Derivative of a Logarithm

Primary Equation: Natural Log (Base e)

$$\frac{d}{dx} \ln(x) = 1/x$$

Secondary Equation: General Base

$$\frac{d}{dx} \log_a(x) = 1 / (x * \ln(a))$$

Domain Restriction

Logarithmic input (x) cannot be negative or zero.

Why? Because a positive base raised to any power will always yield a result greater than zero. Thus, $\ln(0)$ and $\ln(\text{negative})$ are undefined and will crash an ML function.

The Combinatory Toolbox: Expanding Function Complexity

Scalar Multiplication Rule

Rule: Ignore constants attached to variables; they pass through the derivative intact.

$$\frac{d}{dx}(4x^2) \rightarrow 4 * \frac{d}{dx}(x^2) \rightarrow 4 * (2x) = 8x$$

The Sum Rule

Rule: Isolate terms separated by addition or subtraction; derive them independently.

$$\frac{d}{dx}(x + x^2) \rightarrow \frac{d}{dx}(x) + \frac{d}{dx}(x^2) \rightarrow 1 + 2x$$

Insight: Consider the equation of a line $y = mx + c$. The **derivative** of the constant c is **0**. The overall **rate of change** is perfectly equal to its **slope** m .

Handling Intersections: The Product Rule

Used when two separate functions (both dependent on x) are multiplied together.

$$(a * b)' = a * b' + b * a'$$

Step-by-Step Example

$$y = x^2 * \ln(x)$$

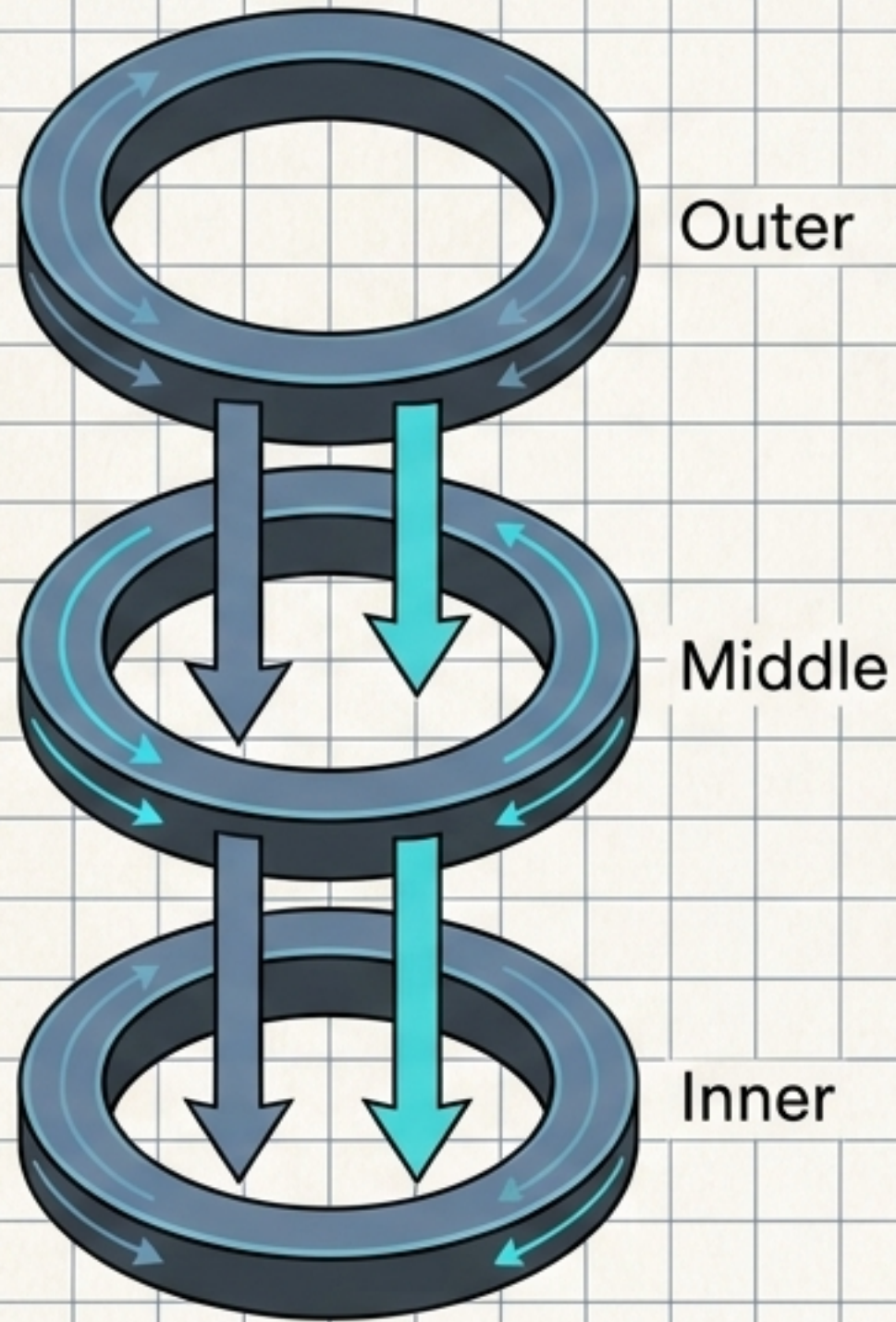
Set $a = x^2$ and $b = \ln(x)$

$$\left(x^2 * \frac{d}{dx} \ln(x) \right) + \left(\ln(x) * \frac{d}{dx} x^2 \right)$$

$$\left(x^2 * \frac{1}{x} \right) + (\ln(x) * 2x)$$

$$\text{Result: } x + 2x \ln(x)$$

Nested Functions: Anatomy of the Chain Rule



The Intuition

When functions are nested inside one another (like a Russian doll), pulling the outer layer affects everything inside it.

The Blueprint

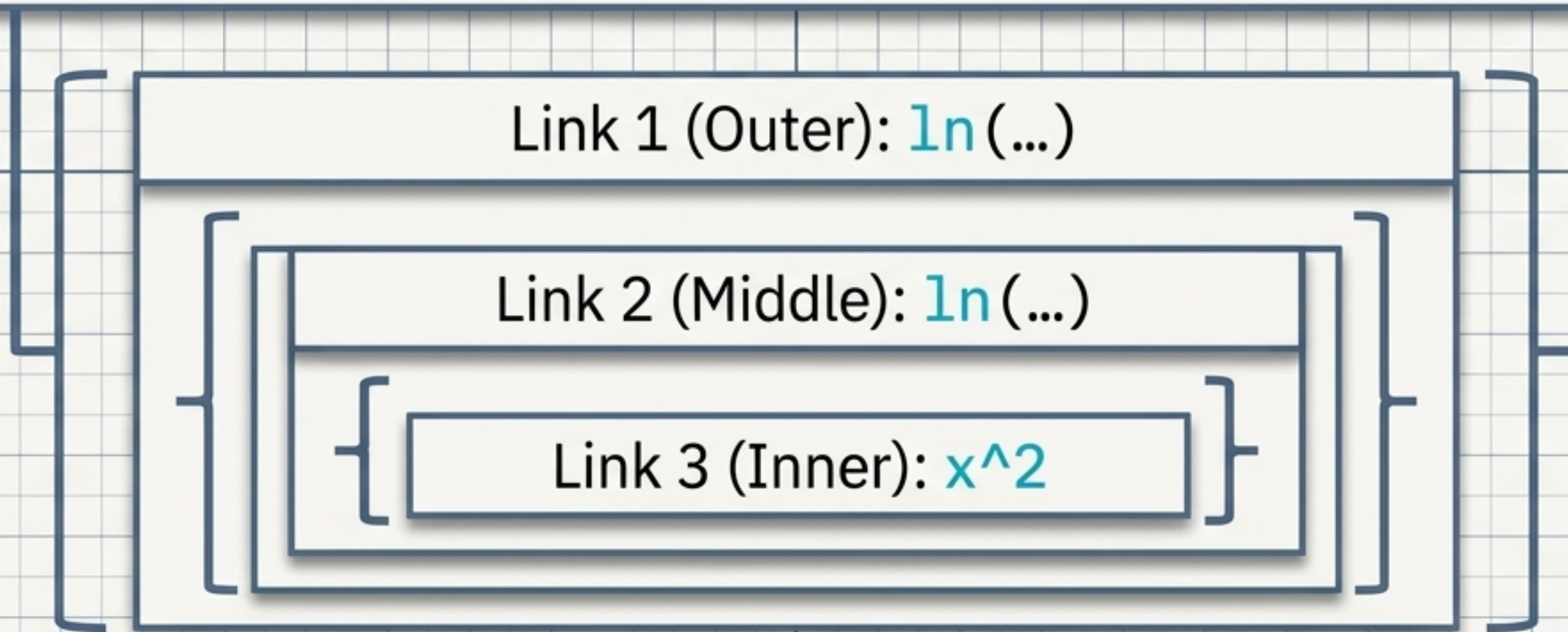
You must take the derivative of the outer function, multiply it by the derivative of the middle function, and multiply that by the derivative of the inner function.

$$\frac{dy}{dx} = \frac{dy}{\cancel{du}} * \frac{\cancel{du}}{dx}$$

The “du” variables mathematically cancel out diagonally, maintaining equation balance.

The Ultimate Test: A 3-Link Chain Rule Execution

$$y = \ln(\ln(x^2))$$



Objective: Calculate the exact rate of change for this intricate nested network.

Step-by-Step Execution

Step 1: Outer Function

Derivative of the outermost $\ln$:

$$\frac{1}{\ln(x^2)}$$

Step 2: Middle Function

Derivative of the inner $\ln$:

$$\frac{1}{x^2}$$

Step 3: Inner Function

Derivative of the core polynomial x^2 :

$$2x$$

Step 4: Assembly & Simplification

$$\frac{1}{\ln(x^2)} * \frac{1}{x^2} * 2x$$
$$\frac{\cancel{2x}}{\cancel{x^2} * \ln(\cancel{x^2})}$$

The ML Tutor's Pro-Tip: An Elegant Shortcut

The Brute Force Result

$$\frac{2x}{x^2 \ln(x^2)} = \frac{2}{x \ln(x^2)}$$

The Magic Shortcut

Logarithmic powers can move to the front:

$$\ln(x^2) = 2 \ln(x)$$

Rewrite the initial equation before deriving:

$$y = \ln(2 \ln(x))$$

$$\text{Result: } \frac{1}{x \ln(x)}$$



Our brute force answer simplifies to exactly this, but this algebraic shortcut completely bypasses the massive chain!

Synthesis: The ML Derivative Cheat Sheet

Section 1: Core Definitions

First Principles

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Section 2: The Rules

$$\text{Power Rule: } \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\text{Natural Log: } \frac{d}{dx} \ln(x) = \frac{1}{x}$$

Section 3: The Operations

Sum Rule:

$$(A+B)' = A' + B'$$

Product Rule:

$$(ab)' = ab' + ba'$$

Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{dx}$$

Master the intuition. Apply the rules. Simplify the code.